

Quick Check

Lesson 8: Why Cell Division Matters: From Baby to Teenager, and Bananas from a Test Tube

9 multiple-choice questions • choose A, B, C or D

Sub-Strand 1.4: Cell Division | Lesson 8

A baby has a body mass of about 3 kg. The same person is 50 kg as a teenager. Which statement best explains what happened to the cells in the body?

- A** The cells only grew bigger, and their number stayed the same.
- B** The number of cells increased because existing cells divided to make new cells.
- C** The food eaten turned directly into new cells without any cell dividing.
- D** The cells joined together to form larger cells.

A human life starts as one fertilised egg cell. Suppose every cell divides into two cells each time, so the numbers go 1, 2, 4, 8, 16 ... How many cells are there after 10 rounds of division?

A 512

B 1024

C 100

D 20

A footballer in Kakamega falls and gets a cut on his knee. After a few days the cut has healed. Which statement best explains how the skin repaired itself?

- A** Skin cells next to the cut divided by meiosis to make new skin cells.
- B** Skin cells next to the cut divided by mitosis to make new, identical skin cells.
- C** The food he ate changed directly into new skin.
- D** The skin cells next to the cut only became larger to fill the gap.

Which type of cell division produces gametes (sex cells) for sexual reproduction and also leads to variation among offspring?

- A** Mitosis
- B** Meiosis
- C** Binary fission
- D** Tissue culture

A tissue-cultured banana plantlet looks exactly like the parent banana plant it came from. What is the best reason?

- A** The plantlet was grown in the same soil as the parent plant.
- B** Its cells were made by repeated mitosis, which produces genetically identical cells.
- C** The nutrient medium in the jar changes the plantlet to look like the parent.
- D** Its cells were made by meiosis, which mixes the genes of both parents.

Tissue-cultured banana plants are all identical to the parent plant. A farmer in Meru worries about this. What is the most sensible concern?

- A** The plants will each divide by meiosis and become different from each other.
- B** The plants will not be able to grow roots.
- C** The plants cannot get their food from the nutrient medium.
- D** Because the plants have the same genes, one disease could affect all of them.

Complete the sentence: 'Food supplies ____, and cell division supplies ____.'

- A** energy only; chromosomes
- B** raw materials and energy; more cells
- C** raw materials; bigger cells only
- D** new cells; raw materials and energy

A group draws the tissue culture side panel of its model poster for tea plants from Kericho. Which order of steps is correct?

A Hardening off, explant, callus, shoots, roots

B Explant, roots, shoots, callus, hardening off

C Callus, explant, roots, shoots, hardening off

D Explant, callus, shoots, roots, hardening off

In a cell that divides by mitosis, what happens to the chromosomes so that the two new cells are genetically identical to the parent cell?

- A** The chromosomes are not copied, so one new cell gets all of them.
- B** The chromosomes are halved, so each new cell gets half of the parent's chromosomes.
- C** The chromosomes are shared out randomly, so each new cell gets a different set.
- D** The chromosomes are copied, and each new cell receives a full set identical to the parent's.

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).